is induced in the secondary winding. This high tension current will reach the respective spark plug as per the firing order through a distributor. Thus, the spark is produced and the air-fuel ,mixture is ignited.

Advantages

1. No moving parts
2. Ignition timing is controlled by ECU, hence no need for an advance mechanism
3. Long life for spark plug
4. Spark intensity is higher
5. Contact breaker points, condenser are eliminated

10.3 Ignition coil

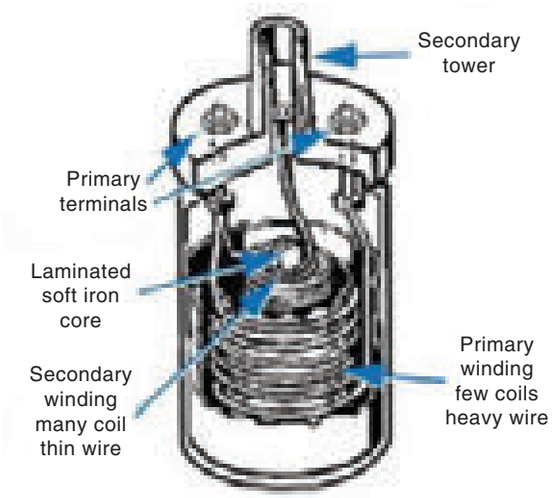


Figure 10.3 (a) Ignition coil

The ignition coil is used to convert 12 V D.C. from the battery to 20,000V to 30,000V A.C. It works on the principle of Faraday's law of electromagnetic induction. It is also called as Step up transformer. The primary coil is made up of thick wires with 150 to 300 turns. The secondary coil is made up of thin wires with 20000 to 25000 turns.

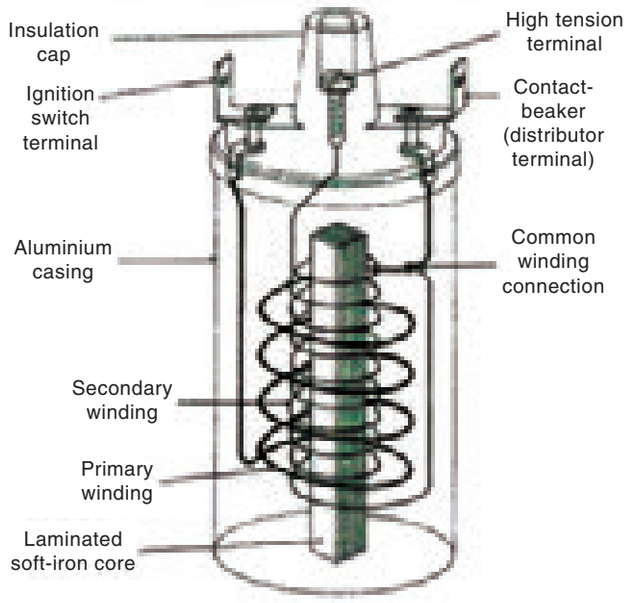


Figure 10.3 (b) Ignition coil

Types of Ignition coil

1. Can type (or) Metal glad type
2. Core type

10.4 Spark plug

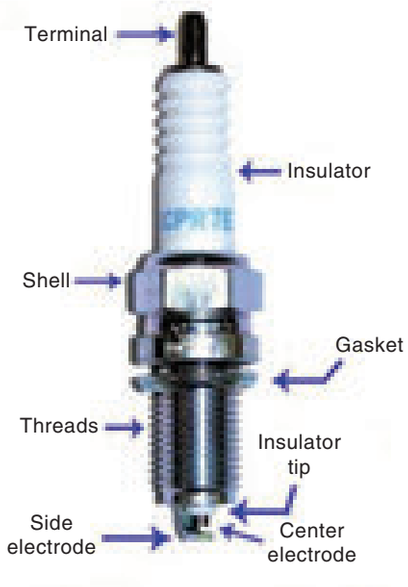


Figure 10.4 (a) Spark plug

In petrol engines, at the end of compression stroke, the spark plug is used to

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